Граници на редици - задачи - отговори

1.

$$\lim_{n\to\infty}\left(\frac{3n^3+n-1}{3n^3-1}\right)^{n^2+2} = \sqrt[3]{e} \tag{1}$$

$$\lim_{n\to\infty}\frac{5n^2+n^3.3^{2n-1}+5e^{2n}}{\operatorname{arctg} n^2+n.4^n+(n!)^{-2}} = \infty \tag{2}$$

$$\lim_{n\to\infty}(n+1+n\cos n)^{(2n+n\cos n)^{-1}} = 1 \tag{3}$$

$$\lim_{n\to\infty} n^3\left(\sqrt{n^2+\sqrt{n^4+1}}-n\sqrt{2}\right) = \frac{1}{4\sqrt{2}} \tag{4}$$

$$\lim_{n\to\infty}\left(\frac{1+\sqrt[n]{a}}{2}\right)^n = \sqrt{a} \tag{5}$$

$$\lim_{n\to\infty}\frac{1}{2^n}\left(\sqrt[n]{\frac{n}{2n+1}}+\sqrt[n]{\frac{n^2}{2n^2+1}}\right)^n = \frac{1}{2} \tag{6}$$

$$\lim_{n\to\infty}\frac{\lg\left(n^2+2n\cos n+1\right)}{1+\ln(n+1)} = \frac{2}{\ln 10} \tag{7}$$

$$\lim_{n\to\infty}\sqrt[n]{\frac{2n^2-5n+3}{n^5+1}} = 1 \tag{8}$$

$$\lim_{n\to\infty}\sqrt[n]{\frac{n^2+4^n}{n+5^n}} = \frac{4}{5} \tag{9}$$

$$\lim_{n\to\infty}\frac{n-\lg n}{\log_2\left(4^n+1\right)} = \frac{1}{2} \tag{10}$$

$$\lim_{n\to\infty}\left(\frac{n+1}{n+2}\right)^{\frac{1-\sqrt{n}}{1-n}} = 1 \tag{11}$$

$$\lim_{n\to\infty}\left(\frac{3n^2-n+1}{2n^2+n+1}\right)^{\frac{n^3}{1-n}} = 0 \tag{12}$$

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{\sqrt{n}}\operatorname{arctg}\left(\frac{n^4+3}{n^4+3\cos^3 n}\right) = \frac{\pi}{4} \tag{13}$$

$$\lim_{n\to\infty}\sqrt[n^2]{\frac{n-1}{n+1}}+\sqrt[n]{3^n n^3+2} = 4 \tag{14}$$

$$\lim_{n\to\infty} n\left(\frac{n+1}{2n-1}\right)^n\cos\left(\frac{n^4+3}{n^7+3\ln^3 n}\right) = 0 \tag{15}$$

$$\lim_{n\to\infty}\frac{n\sqrt[n]{\lg n}}{\left(1+\frac{1}{n}\right)^{n^2}} = 0 \tag{16}$$

$$\lim_{n\to\infty}\frac{\lg\left(5n^2+3n+1\right)}{\sqrt{n}+1} = 0 \tag{17}$$

$$\lim_{n\to\infty}\frac{\operatorname{arccotg}\left(-n^2+3n+1\right)}{\arcsin\frac{n+1}{n+2}} = 2 \tag{18}$$

$$\lim_{n\to\infty}\frac{\arccos\left(\frac{2n+3}{n^3+3}\right)}{(n^2-3)\ln\frac{n^2+1}{n^2+2}} = -\frac{\pi}{2} \tag{19}$$

2.

$$\lim_{n\to\infty} \frac{1}{\sqrt[n]{n!}} = 0 \tag{20}$$

$$\lim_{n\to\infty} \frac{n!}{(2n+1)!!} = 0 \tag{21}$$

$$\lim_{n\to\infty} \frac{(2n-1)!!}{(2n)!!} = 0 \tag{22}$$

$$\lim_{n\to\infty} \frac{\sqrt[n]{n!}}{n} = \frac{1}{e} \tag{23}$$

$$\lim_{n\to\infty} \frac{4^n n!}{(3n)^n} = 0 \tag{24}$$

$$\lim_{n\to\infty} \frac{3^n n!}{n^n} = \infty \tag{25}$$

$$\lim_{n\to\infty} \sqrt[n]{(n+1)(n+2)...(2n)} = \infty \tag{26}$$